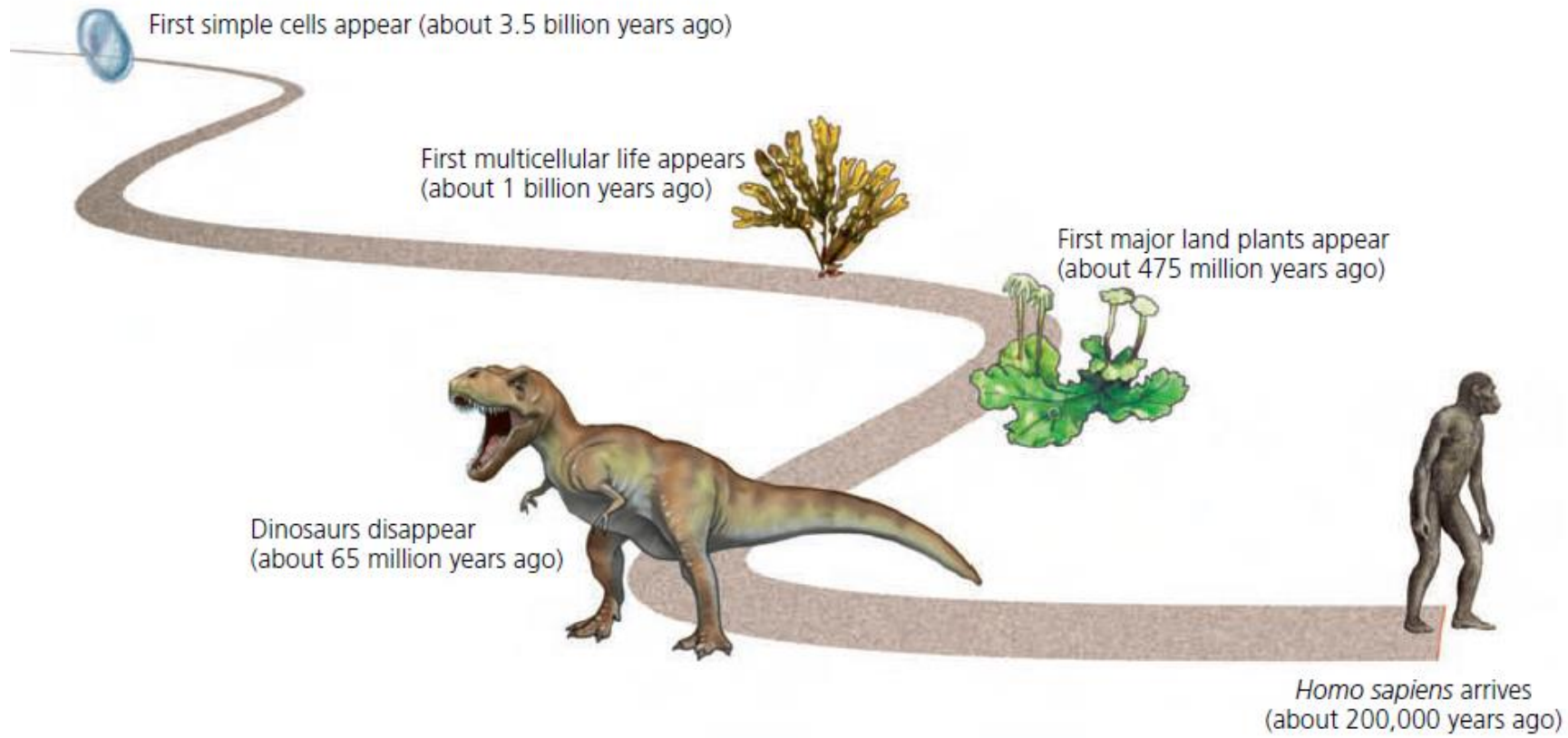


Biodiversity

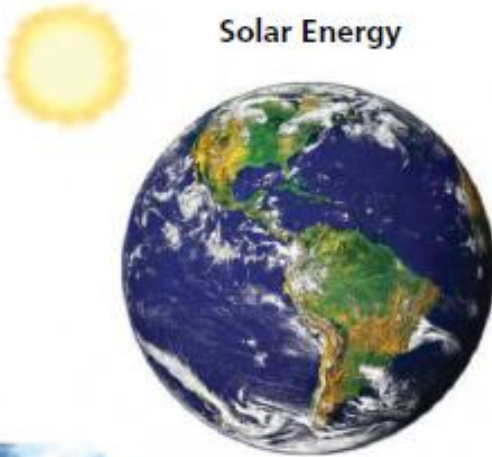


Sustainability

- Considering the billions of years that life has existed on the earth, our species has been around for less than the blink of an eye
- Within only a few hundred years, we have taken over most of the earth to support our basic needs and rapidly growing wants. But in the process, we have degraded much of the earth.

Sustainability

Solar Energy



The following are called three principle of sustainability

1. Solar energy,
2. Biodiversity
3. Chemical Recycling

They helps the existence of life on this planet over billions of years.



Chemical Cycling



Biodiversity

Sustainability

There are three overarching themes relating to the long-term sustainability of life on this planet

- *solar energy- rely on the sun*
- *Biodiversity-promote multiple options for life*
- *chemical cycling-reduce wastage*

Biodiversity

- Importance
- types
- mega-biodiversity
- Species interaction
- Extinct, endemic, endangered and rare species;

Biodiversity

- **Biological diversity**, or **biodiversity**, is the variability among living organisms from all sources including *inter alia*, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are a part.

Types of biodiversity

Types of Biodiversity

- *Species diversity*
- *Genetic diversity*
- *Ecosystem diversity*
- *Functional diversity*

Species diversity

- **Species diversity** is the variety of species per unit area and this include both the **species richness** (number of species present) and their **relative abundance** (how many of each species per unit area)

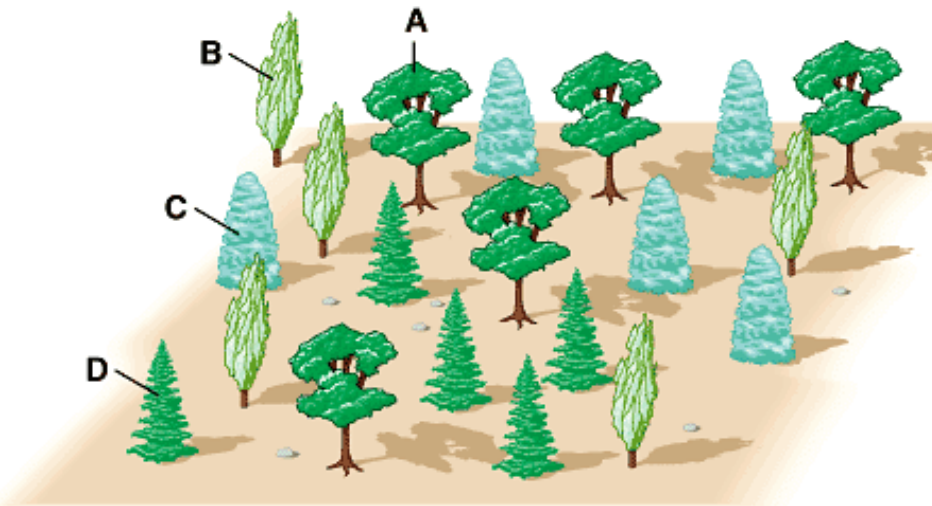
Species diversity

- We do not know how many species there are on the earth.
- Estimates range from 8 million to 100 million.
- So far, biologists have identified about 1.9 million

Species Diversity

A large group of diverse animals, including giraffes, elephants, lions, tigers, zebras, rhinos, hippos, camels, and birds, standing together on a grassy plain under a cloudy sky. The animals are arranged in a dense cluster, showcasing a wide variety of species. The background features a bright blue sky with scattered white clouds, and the foreground is a lush green field. The overall scene conveys a sense of biodiversity and natural harmony.

Species Diversity



Community 1

A: 25% B: 25% C: 25% D: 25%



Community 2

A: 80% B: 5% C: 5% D: 10%

In these images we can see that the two communities are composed of the *same* species richness, but they *differ* in regards to species abundance:

Genetic diversity

Genes are the basic units of hereditary information transmitted from one generation to other

The genes found in **organisms** (e.g. Plants, animals, humans) can form enormous number of combinations each of which gives rise to some variability.

When the genes within the same **species** (e.g. cats, lions) show different versions due to new combinations, it is called genetic diversity.

Genetic diversity



Genetic Diversity in Human



Ecosystem diversity

- ***Variation in the habitat*** in which species live and interact
- ***Different types*** of forest, grass lands, wet lands, water bodies like streams, lakes and oceans, mountains...



Functional diversity

The variety of processes such as energy flow and matter cycling that occur within ecosystems as species interact with one another in food chains and food webs.

Importance of biodiversity

Importance of biodiversity

- The earth's biodiversity is a vital part of the natural capital that helps keep us alive
- Biodiversity also plays critical roles in preserving
 - a) quality of the air and water
 - b) maintaining the fertility of topsoil
 - c) Decomposing and recycling waste
 - d) controlling populations of species that humans consider to be pests.

Biodiversity helps to sustain life on the earth.

Importance of Biodiversity

Direct Values

- Direct values are concerned with the enjoyment or satisfaction received *directly* from biological resources.
- Relatively *easily observed and measured* by assigning prices to them.
- There are two types including *Consumptive* and *Productive* use.

Importance of Biodiversity – Direct values

Consumptive use

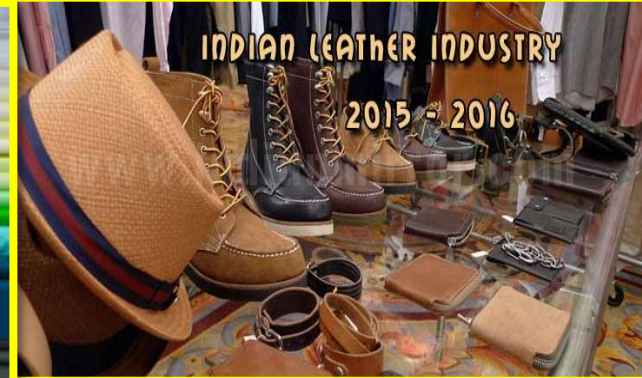
- Direct values where the biodiversity products can be harvested and consumed directly, like *food, fuel, milk, fiber etc.*
- Humans are mostly dependent on plants and animal resources for *dietary requirements*. A major share of the food comes from domesticated crops and animals.
- A significant percentage of world population depends upon plants or plant extracts for *medicines*.
- Forests have been used since ages for *fuel wood*. Firewood collected by individual is not normally marketed, but is directly consumed by *tribes and local villagers*.

Direct Values – Consumptive Use



Importance of Biodiversity – Directive values

- **Productive values** are commercially usable values where the product is marketed and sold (Productive value is assigned to products that are ***commercially harvested and marketed***).
- **Animal products** like tusks of elephants, musk from musk deer, silk from silkworm, wool from sheep... all of which are traded in the market.
- Several industries dependent upon the productive use values of biodiversity, like ***paper industry, plywood industry, textile industry, leather industry, pearl industry etc.***



Direct Values – Productive Use



Importance of Biodiversity-indirect values

- These are **indirect values** of ecosystem functions.
- **Carbon fixing** through photosynthesis, which provides the **support system** for species.
- **Pollination**
- Maintaining **nutrient cycles**
- Soil **production and protection** from erosion
- **Absorbing and decomposing** pollutants, organic wastes, pesticides, air and water pollutants
- **Regulating climate**
- Preserving recreational, aesthetic, cultural and historic values of **natural environment**

Importance of Biodiversity

- **Non-consumptive values** deal primarily with the functions of ecosystems.
- *Do not normally appear* in national accounting systems, but they may **outweigh** direct values when they are computed.
- Non-consumptive values include
 - Social values**
 - Aesthetic values**
 - Ethical values**

Endemic Species

Endemic Species

- Species that are found in only one area are called **endemic species** and are especially vulnerable to extinction.



Bicolored frog
Western Ghat, India



Lemur
Madagascar



Giant Panda
China

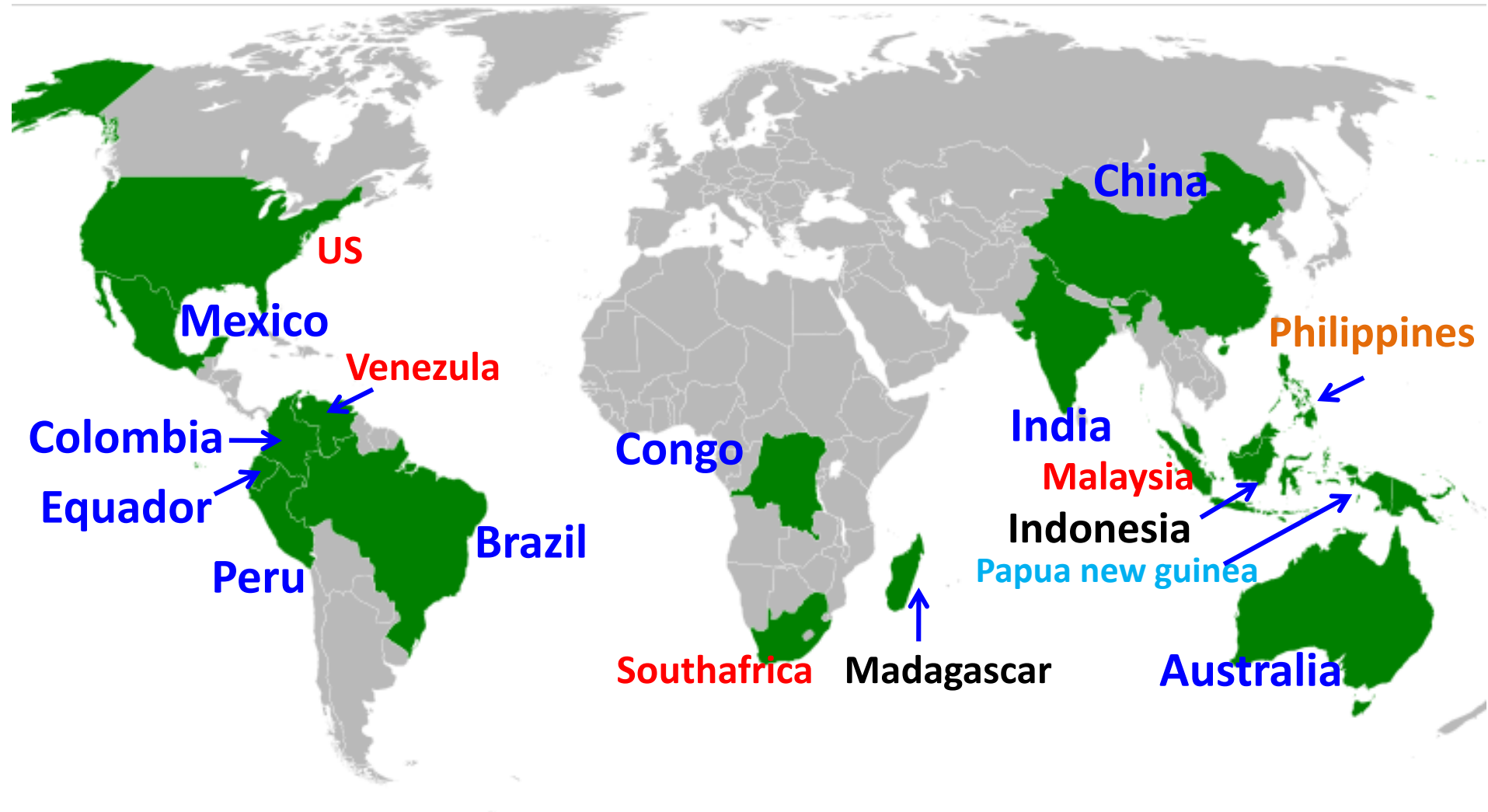
Megadiversity

Megadiversity

- **Megadiverse country** refers to any one of a group of nations that harbour the *majority of earth's species and high number of endemic species*.
- Megadiversity means exhibiting **great diversity**.
- Main criteria for megadiverse countries is **endemism** at the level of species, genera and families.
- (**Endemic species** are plants or animals which are native to and restricted to a particular geographical region)
- A megadiverse country must have at least **5,000 species** of endemic plants and must border marine ecosystems.

Megadiversity

There are **17** Megadiverse countries.



Species Interaction

Species Interaction

Ecologists identify five basic types of interactions between species as they share limited resources such as food, shelter, and space:

- **Interspecific competition**
- **Predation**
- **Parasitism**
- **Mutualism**
- **Commensalism**

Species Interaction

- **Interspecific competition**

occurs when members of two or more species interact to gain access to the same limited resources such as food, water, light, and space.



Species Interaction

- Interspecific competition

Each species plays a role in its ecosystem called its *ecological niche*

What is ecological niche

It is a species' way of life in a community and includes everything that affects its survival and reproduction, such as how much water and sunlight it needs, how much space it requires, what it feeds on, what feeds on it, and the temperatures it can tolerate.

Species Interaction

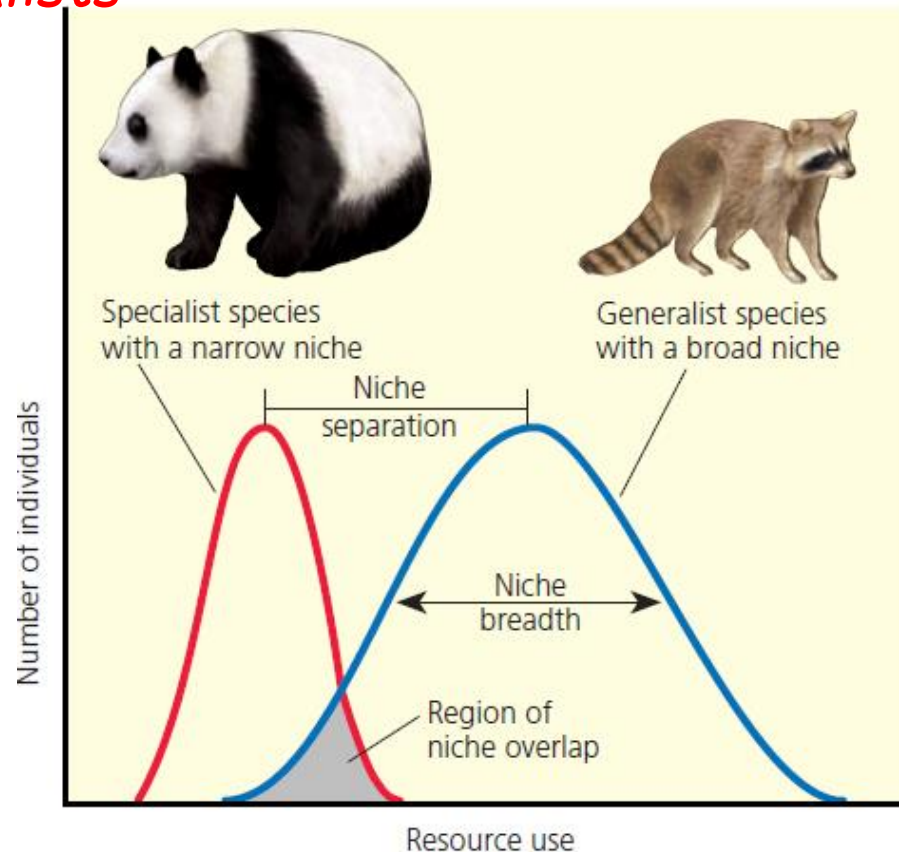
- Interspecific competition

Each species plays a role in its ecosystem called its *ecological niche*

Scientists use the *niches* of species to classify them broadly as *generalists or specialists*

Generalist species have broad niches

specialist species occupy narrow niches



Species Interaction

- Interspecific competition

Each species plays a role in its ecosystem called its *ecological niche*

Some species are specialists with a narrow niche and some are generalists with a broad niche.

When two species compete with one another for the same resources such as food, light, or space, their *niches overlap*.

The greater this overlap, the more intense their competition for key resources.

Species Interaction

Predation: Occurs when a member of one species (the predator) feeds directly on all or part of a living organism (the prey).



Species Interaction

- **Parasitism**

Parasitism occurs when one species (the *parasite*) feeds on another organism (the *host*), usually by living on or inside the host. In this relationship, the parasite benefits and the host is often harmed but not immediately killed.

Examples: Bedbugs, tapeworms, mosquito



Species Interaction

- **Parasitism**

From the host's point of view, parasites are harmful.

But from the population perspective, parasites can promote biodiversity by contributing to species richness,

in some cases, they help to keep the populations of their hosts in check.

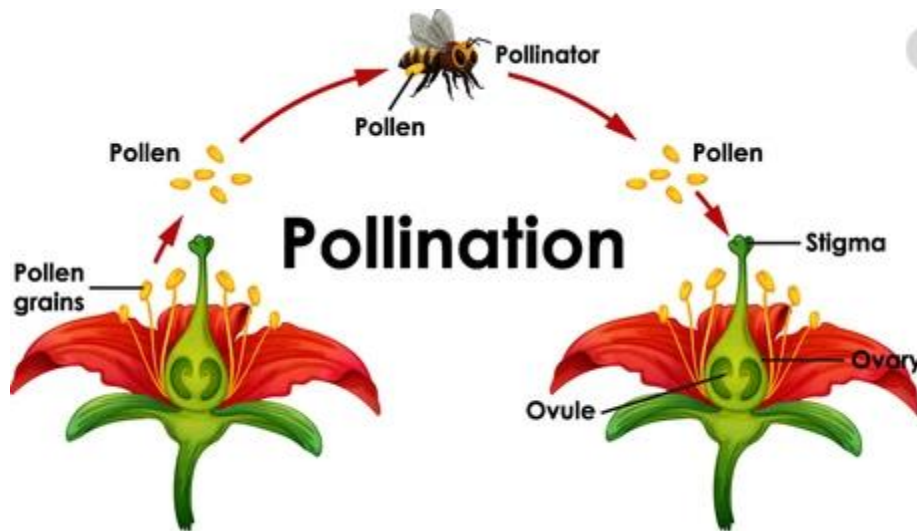
Species Interaction

- **Mutualism**

In **mutualism**, two species behave in ways that benefit both by providing each with food, shelter, or some other resource.

pollination of a flower by species such as honeybees

butterflies that feed on the flower's nectar



Species Interaction

- **Mutualism**



(a) Oxpeckers and black rhinoceros

mutualistic relationships that combine *nutrition* and *protection*.

The birds remove and eat parasites and pests (such as ticks and flies) from the animal's body and often make noises warning the larger animals when predators are approaching.

Species Interaction

- Commensalism

Commensalism is an interaction that benefits one species but has little, if any, beneficial or harmful effect on the other.



birds benefit by nesting in trees,
generally without harming them.

Species Interaction

- Commensalism



plants called *epiphytes* (such as certain types of orchids and bromeliads), which attach themselves to the trunks or branches of large trees in tropical and subtropical forests

Species Extinction

Species Extinction

The balance between formation of new species and extinction of existing species determines the earth's **biodiversity**.

A species is said to be **extinct** when it is not seen in the wild for 50 years at a stretch e.g. passenger pigeon.

Effect on ecosystem

Biological extinction of certain important species can **weaken or break** some of the connections in the **ecosystem**

Biological extinction can lead to **secondary extinctions (co-extinction)** of species with strong connections to species that have already gone extinct.

Species Extinction

Species extinction can happen naturally

During most of the 3.5 billion years that life has existed on the earth, there has been a natural, low rate of species extinction known as the **background extinction rate**.

The extinction of many species in a relatively short period of geologic time is called a **mass extinction**.

earth has experienced mass extinctions. After each mass extinction, biodiversity eventually returned to equal or higher levels, but each recovery required millions of years.

Species Extinction

Species extinction can happen naturally

Reasons for species extinction

Major climate change

Volcanic eruption

a large asteroid hit the planet

Species Extinction

Some Human Activities are Causing Extinctions

Extinction is a natural process but it has accelerated as human populations have spread over most of the globe,

We consume huge quantities of resources and creating large and growing ecological footprints

filling in wetlands or converting grasslands and forests to crop fields and urban areas.

Human activities have also polluted and disturbed almost half of the surface waters

Species Extinction

Some Human Activities are Causing Extinctions

- Current annual rate of species extinction is up to 1,000 times the historical background rate of about 0.0001%.
- biodiversity losses are likely to increase sharply during the next 50-100 years because of the projected growth of the human population and its growing **use of resources per person**,
- we may be **limiting the long-term recovery** of biodiversity by reducing the rate of speciation for some species. In other words, we are also creating a speciation crisis.

Some of the extinct species



Passenger Pigeon
Extinction in 1914



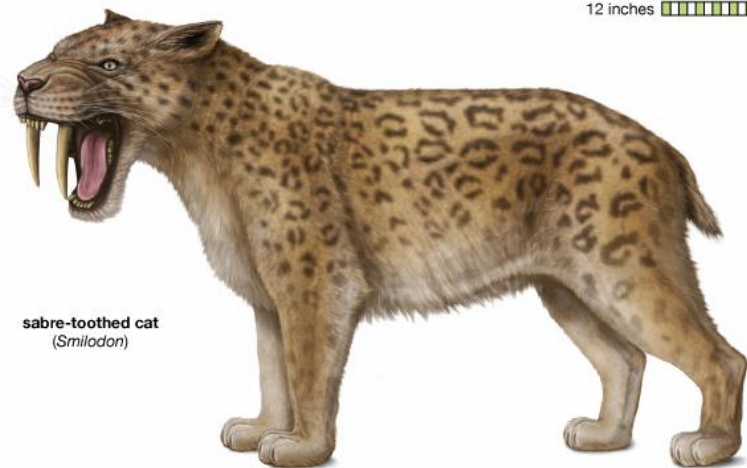
Tasmanian Tiger
Extinction in 1936



Dodo
Extinction in 1662



Mammoth
Extinction 10,000 – 4000 BC



Sabre-toothed cat
Extinction about 10,000 BC

Passenger Pigeon Extinction



- These birds were good to eat,
- Their feathers made good pillows,
- Their bones were widely used for fertilizer.

North America's passenger pigeon extincted because

- habitat loss
- uncontrolled commercial hunting
- They are easy to kill

Endangered Species

Endangered Species

Biologists classify species that are heading toward biological extinction as

Endangered

Threatened

rare

An **endangered species** has so few individual survivors that the species could soon become extinct.

A **threatened species** (also known as a *vulnerable species*) still has enough remaining individuals to survive in the short term, but because of declining numbers, it is likely to become endangered in the near future.

Endangered Species

Biologists classify species that are heading toward biological extinction as

Endangered

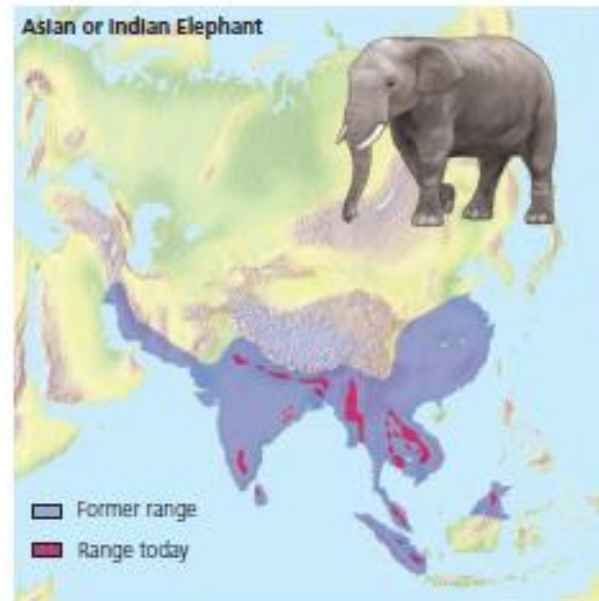
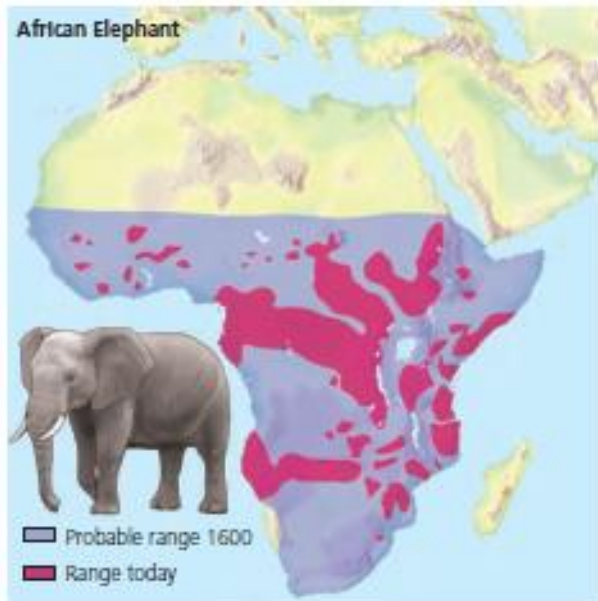
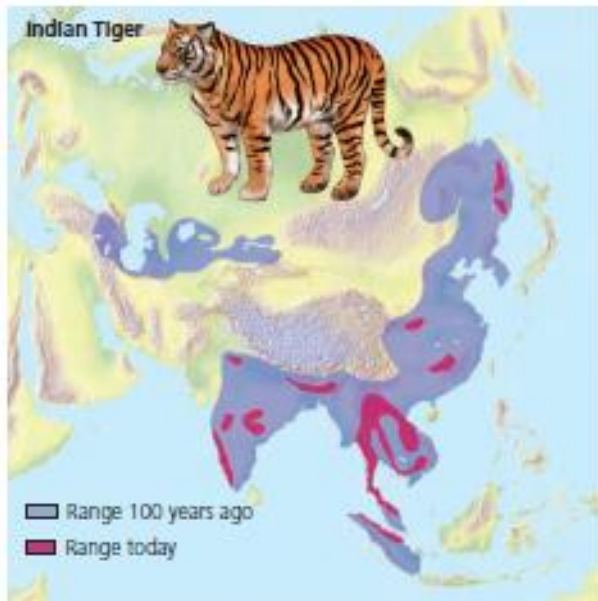
Threatened

rare

Species which are not endangered or vulnerable at present, but are at a risk are categorized as **rare species**.

they are usually endemic. Sometimes they are thinly scattered over a more extensive area.

Endangered Species



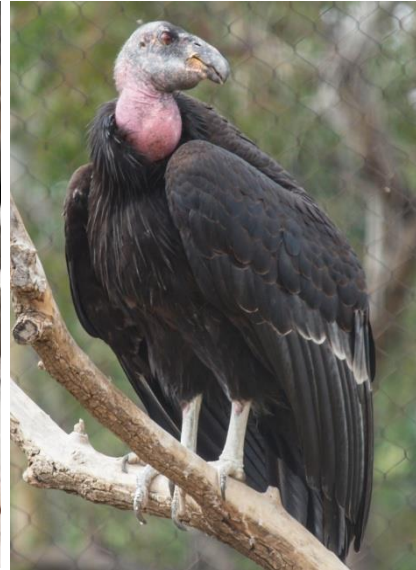
Some of the endangered species



Sumatran Tiger
<60 in Indonesia



Mexican Grey Wolf
< 60 in New Mexico



California condor
172 in US



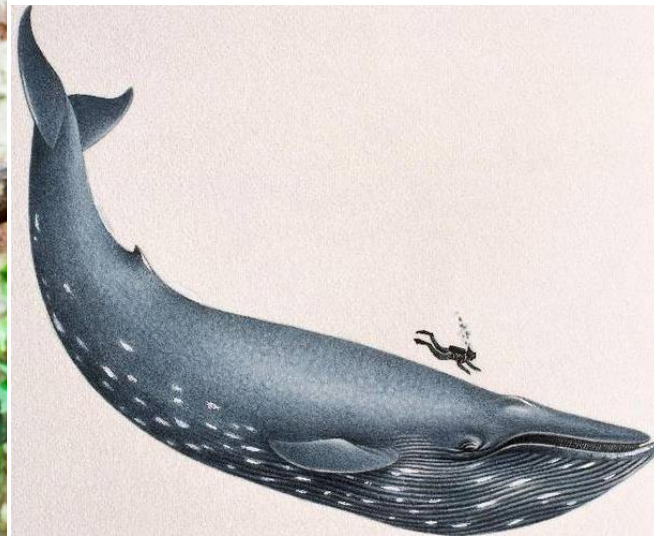
Whooping Crane
210 in US



Indian Bustard



Orangutan



Blue Whale

Endemic Species

- They exist on islands and in other unique areas, especially in tropical rain forests where most species have highly specialized roles.

Extinction of Endemic Species is because

- Habitat loss
- changing environmental conditions

Endemic Species



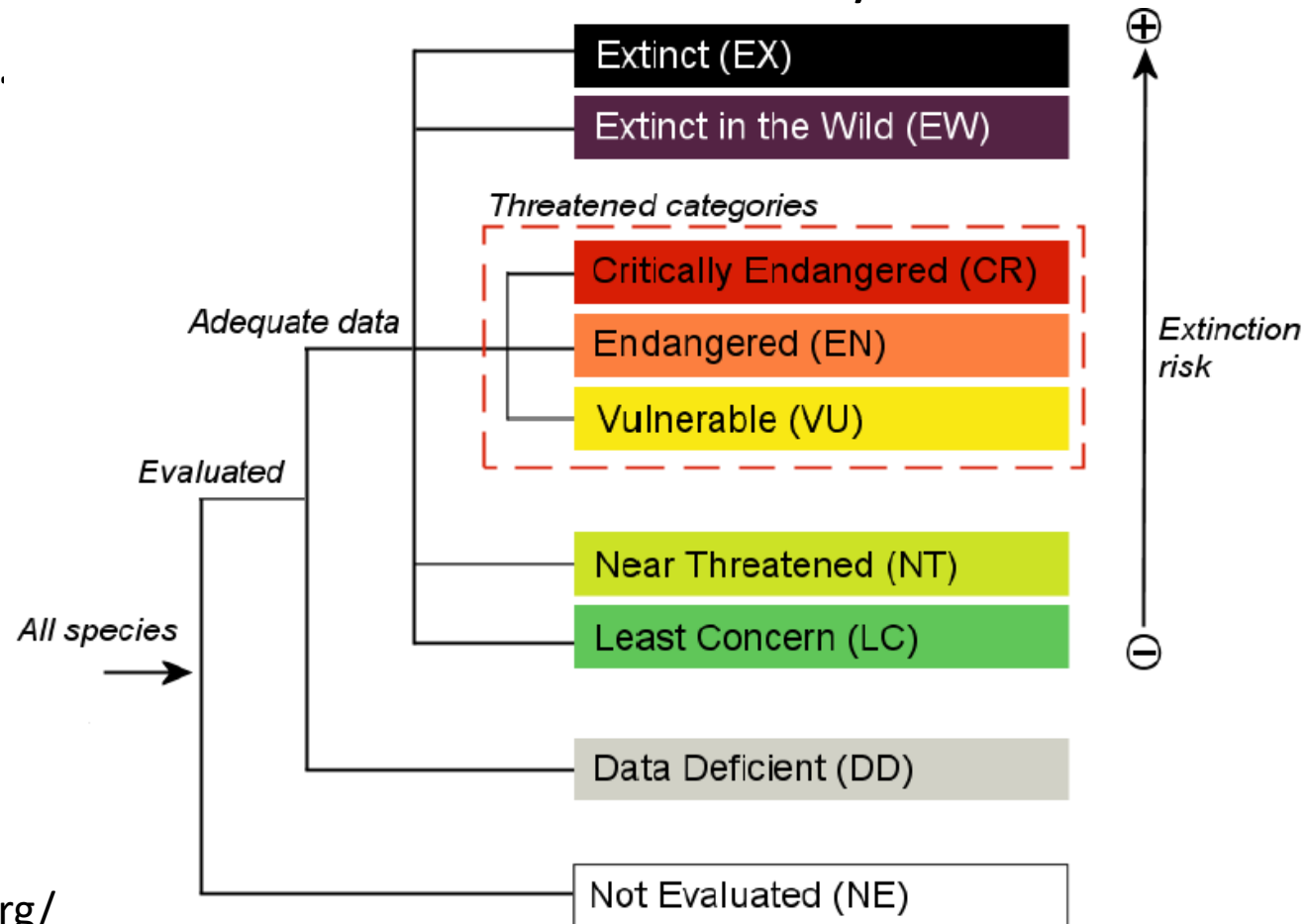
Michael P. Fogden/Bruce Coleman USA

This golden toad lived in Costa Rica's high-altitude Monteverde Cloud Forest Reserve.

The species became extinct in 1989 apparently because its habitat dried up.

IUCN Red list

- IUCN (International Union for Conservation of Nature) evaluates the conservation status of plants and animal species.
- IUCN was set up in 1948 and the IUCN Red list system was first conceived in 1963.

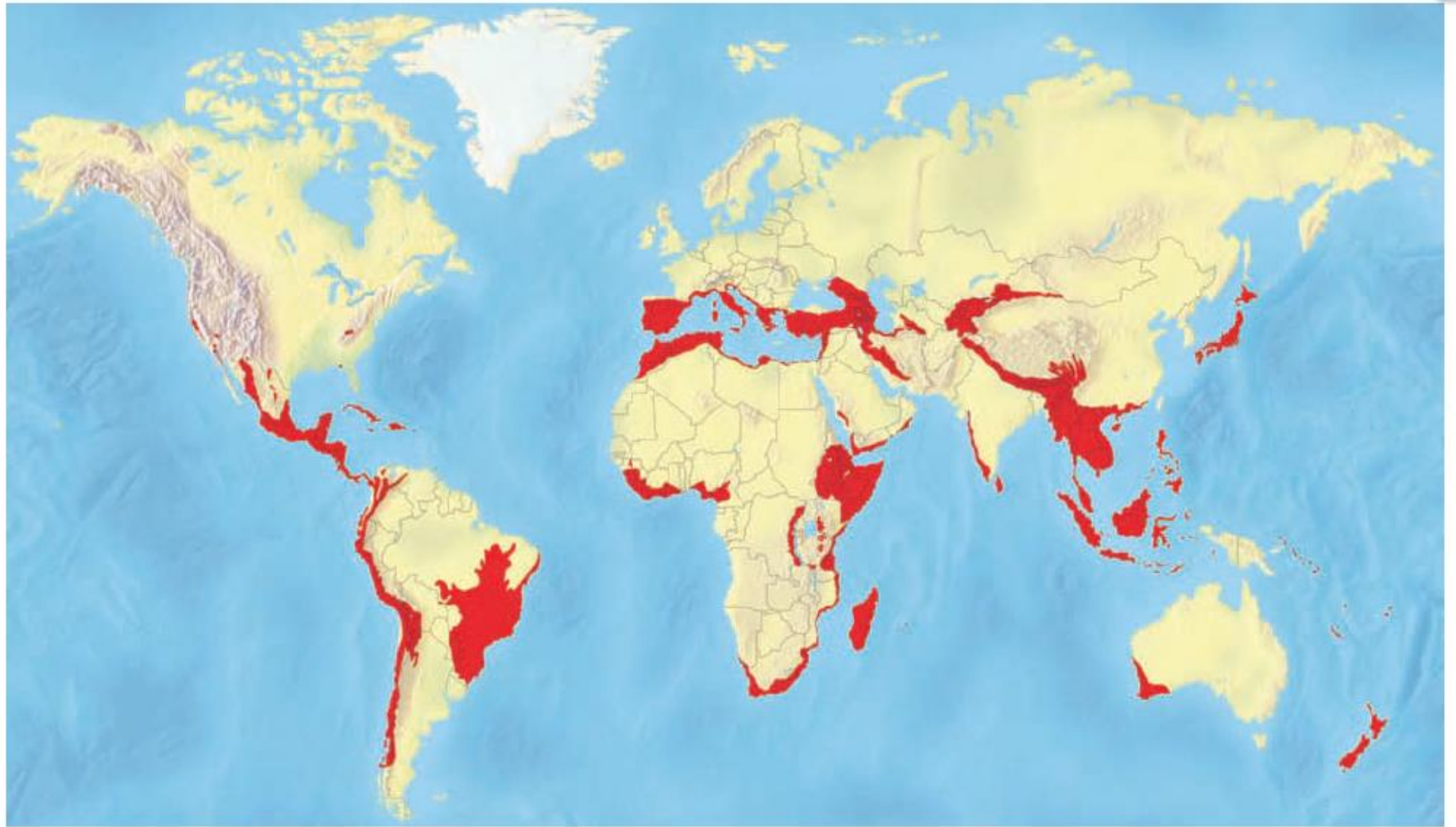


Hotspots

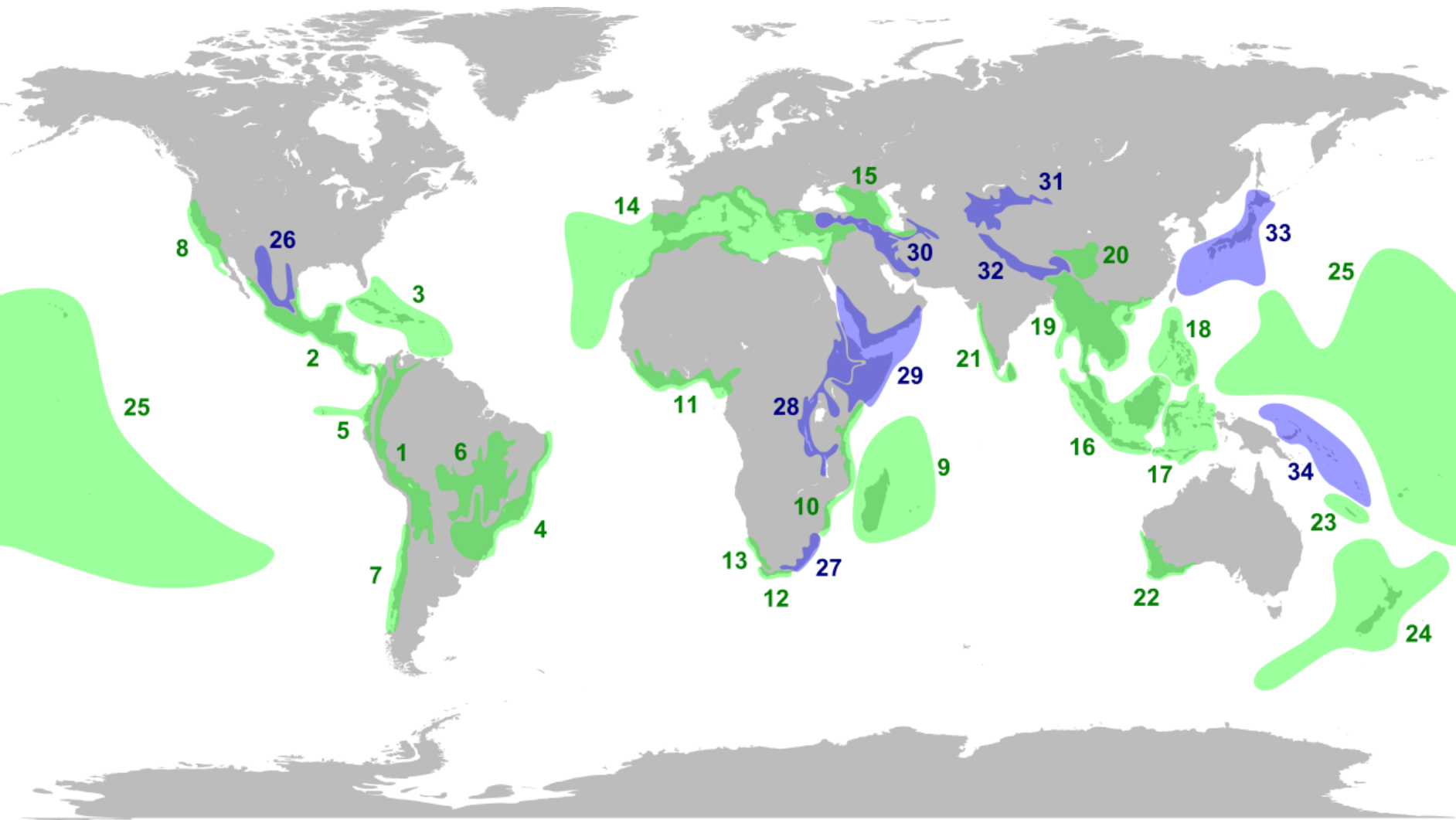
Biodiversity Hotspots

- Biogeographical region with significant levels of biodiversity that is ***threatened with destruction***.
- To qualify as a biodiversity hotspot, the region must
 - a) Contains at least **0.5% or 1500 species** of vascular plants as endemic, and
 - b) Has to have ***lost*** at least **70%** of its primary vegetation.
- Around the world, **36** areas considered as biodiversity hotspots.
- These 36 regions support nearly **60%** of the world's plant, bird, mammal, reptile and amphibian species.

Biodiversity Hotspots



36 Biodiversity Hotspots in World



*Original Proposal are in **green** and added regions are in **blue***

- **North and Central America – 5**

- California Floristic Province (8)
- Madrean pine-oak woodlands (26)
- Mesoamerica (2)
- Caribbean islands (3)
- North american coastal plane (36)

- **South America - 5**

- Atlantic forest (4)
- Cerrado (6)
- Chilean winter rainfall- valdivian forest (7)
- Tumbes-choco-magdalena (5)
- Tropical andes (1)

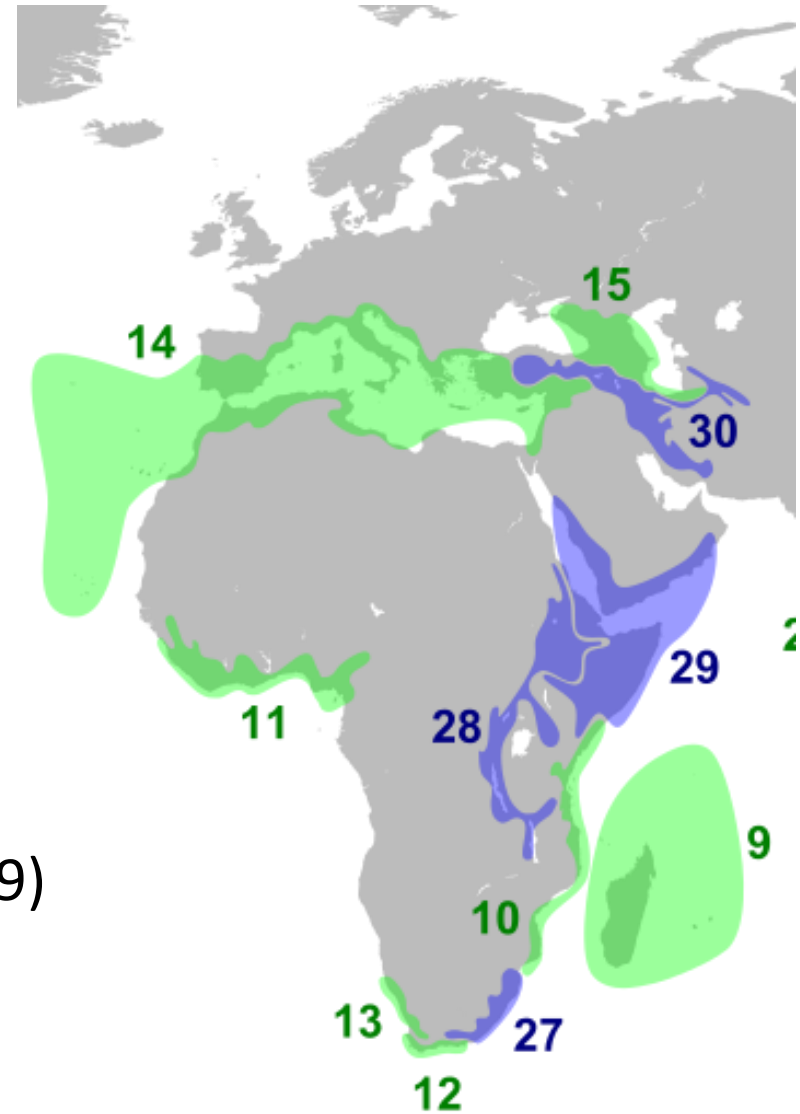


- **Europe - 1**

- Mediterranean basin (14)

- **Africa – 8**

- Cape floristic region (12)
- Coastal forests of eastern africa (10)
- Eastern afromontane (28)
- Guinean forest of west africa (11)
- Horn of africa (29)
- Madagascar and Indian ocean island (9)
- Maputaland-pondoland-albany (27)
- Succulent Karoo (13)



- **Central Asia – 1**

- Mountains of central asia (31)

- **South Asia – 3**

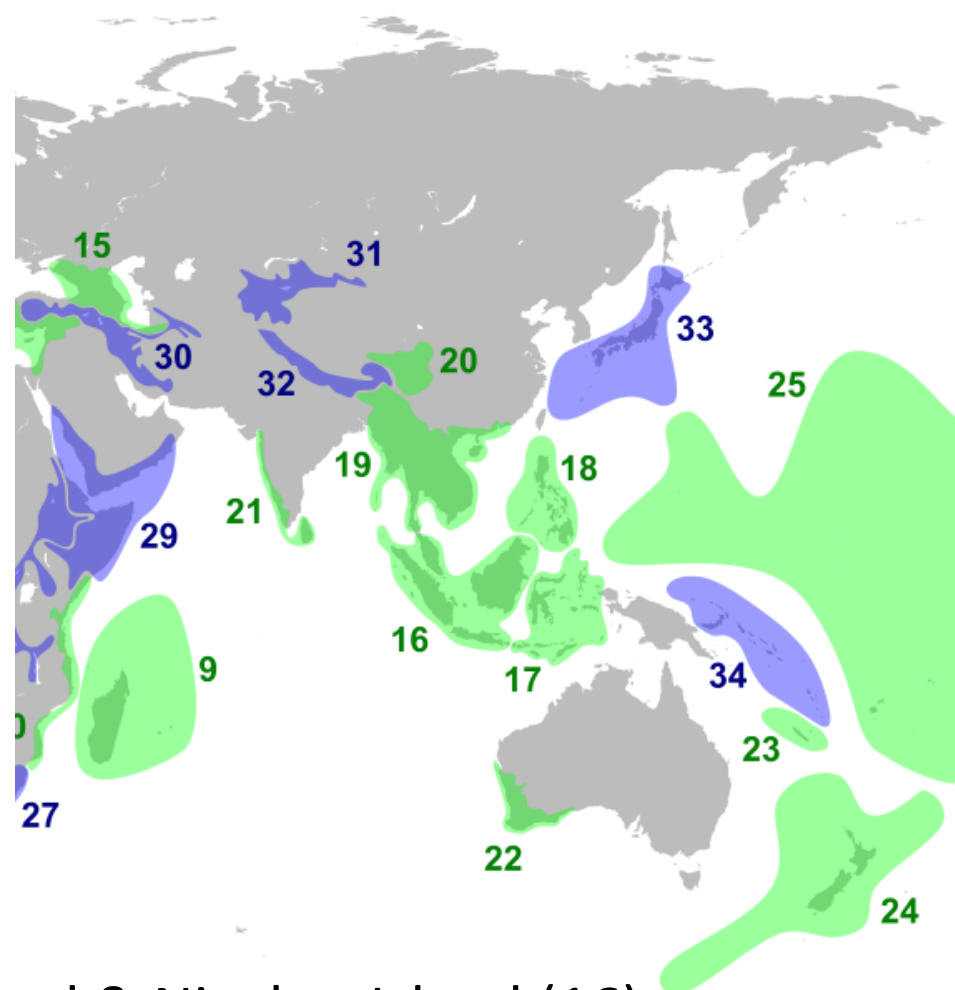
- Eastern Himalaya (32)
- Indo-Burma, Myanmar (19)
- Western Ghats (21)
- Sri Lanka (21)

- **South East Asia, Asia Pacific –9**

- East Melanesian Island (34)
- New Caledonia (23)
- New Zealand (24)
- Philippines (18)
- Polynesia-Micronesia (25)
- Southwest Australia (22)
- Eastern Australian temperate forest
- Sundaland & Nicobar Island (16)
- Wallacea (17)

- **East Asia – 2** (Japan, Mt. Southwest china)

- **West Asia - 2** (Caucasus, Irano-Anatolian)



Biodiversity

- Importance
- types
- mega-biodiversity
- Species interaction
- Extinct, endangered and rare species;

Threats to biodiversity

Threats to biodiversity

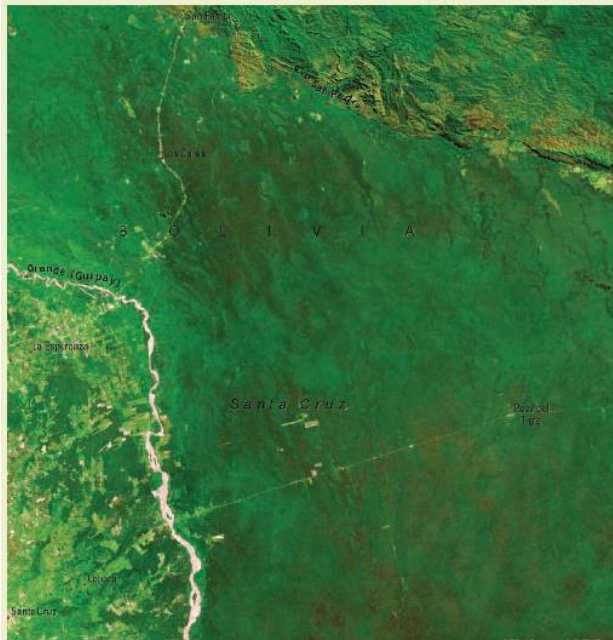
- Habitat loss
- Habitat degradation and fragmentation
- Introduction of non-native species
- Commercial hunting and poaching
- Sale of exotic pets and decorative plants
- Predator and pest control
- Pollution
- Climate change
- Overfishing

Threats to biodiversity

Loss of Habitat Is the Single Greatest Threat to Species:

According to biodiversity researchers, the greatest threat to wild species is habitat loss, degradation, and fragmentation.

Bolivian city of Santa Cruz



1975



2003

Threats to biodiversity

Habitat fragmentation occurs when a large, intact area of habitat such as a forest or natural grassland is divided, typically by roads, logging operations, crop fields, and urban development, into smaller, isolated patches or "habitat islands"

affect

It can divide populations of a species into smaller, increasingly isolated groups that are more vulnerable

Threats to biodiversity

Loss of Habitat Is the Single Greatest Threat to Species:

- Deforestation in tropical areas
- destruction and degradation of coral reefs
- Degradation of coastal wetlands,
- the plowing of grasslands
- the pollution of streams, lakes, and oceans.



Threats to biodiversity

Some Deliberately Introduced Species Can Disrupt Ecosystems

introduced species such as corn, wheat, rice, and other food crops, as well as species of cattle, poultry, and other livestock

affect

In their new habitats, some introduced species face no natural predators, competitors, parasites, or pathogens that would help to control their numbers in their original habitats.

Such non-native species can thus crowd out populations of many native species, trigger ecological disruptions, cause human health problems

Threats to biodiversity

Some Accidentally Introduced Species Can Disrupt Ecosystems

- Many unwanted non-native invaders arrive from other continents as stowaways on aircraft, in the ballast water of tankers and cargo ships

Affects

- Many tourists return home with living plants that can multiply and become invasive.
- Some of these plants might also contain insects that can escape, multiply rapidly, and threaten crops.

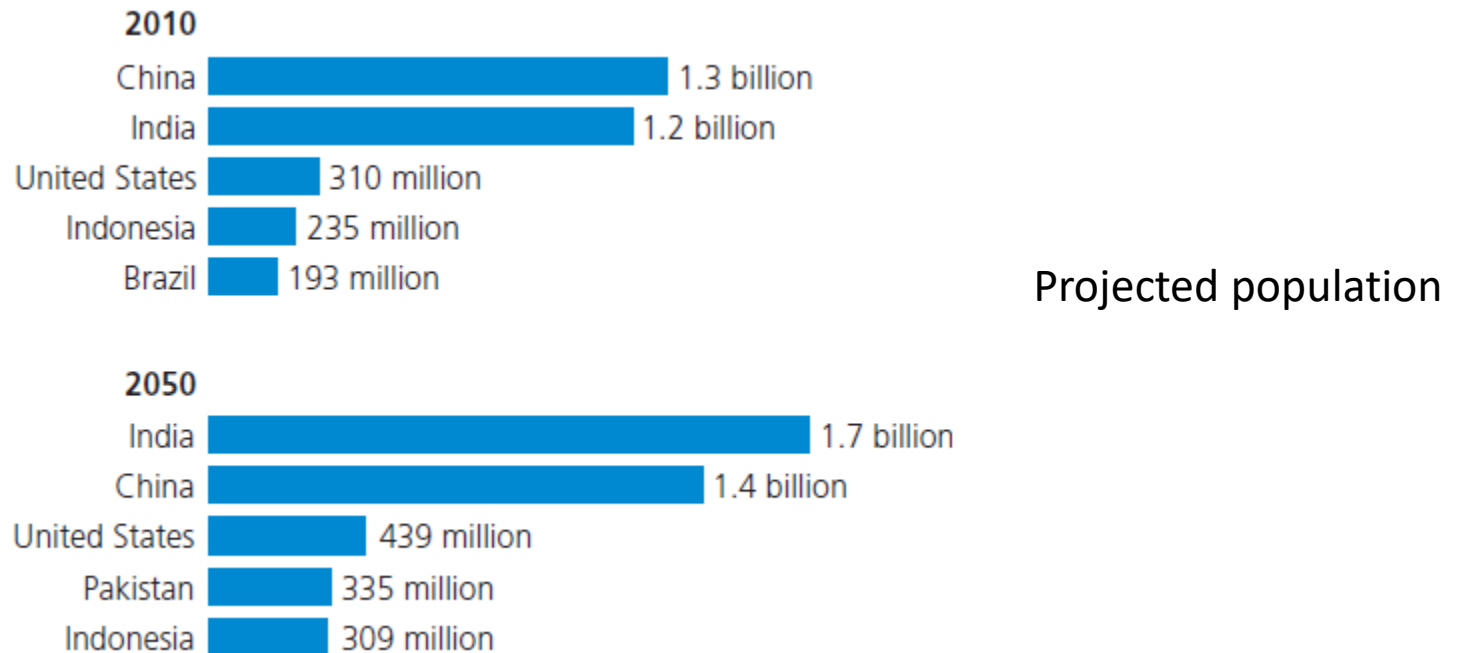
What Can You Do?

Controlling Invasive Species

- Do not capture or buy wild plants and animals.
- Do not remove wild plants from their natural areas.
- Do not release wild pets back into nature.
- Do not dump the contents of an aquarium into waterways, wetlands, or storm drains.
- When camping, use wood found near your campsite instead of bringing firewood from somewhere else.
- Do not dump unused bait into waterways.
- After dogs visit woods or the water, brush them before taking them home.
- After each use, clean your mountain bike, canoe, boat, motor, and trailer, all fishing tackle, hiking boots, and other gear before heading for home.

Threats to biodiversity

Population Growth and Overconsumption



human population growth and excessive and wasteful consumption of resources have greatly expanded the human ecological footprint,

ecological footprint has eliminated, degraded, and fragmented vast areas of wildlife habitat

Threats to biodiversity

- Pollution

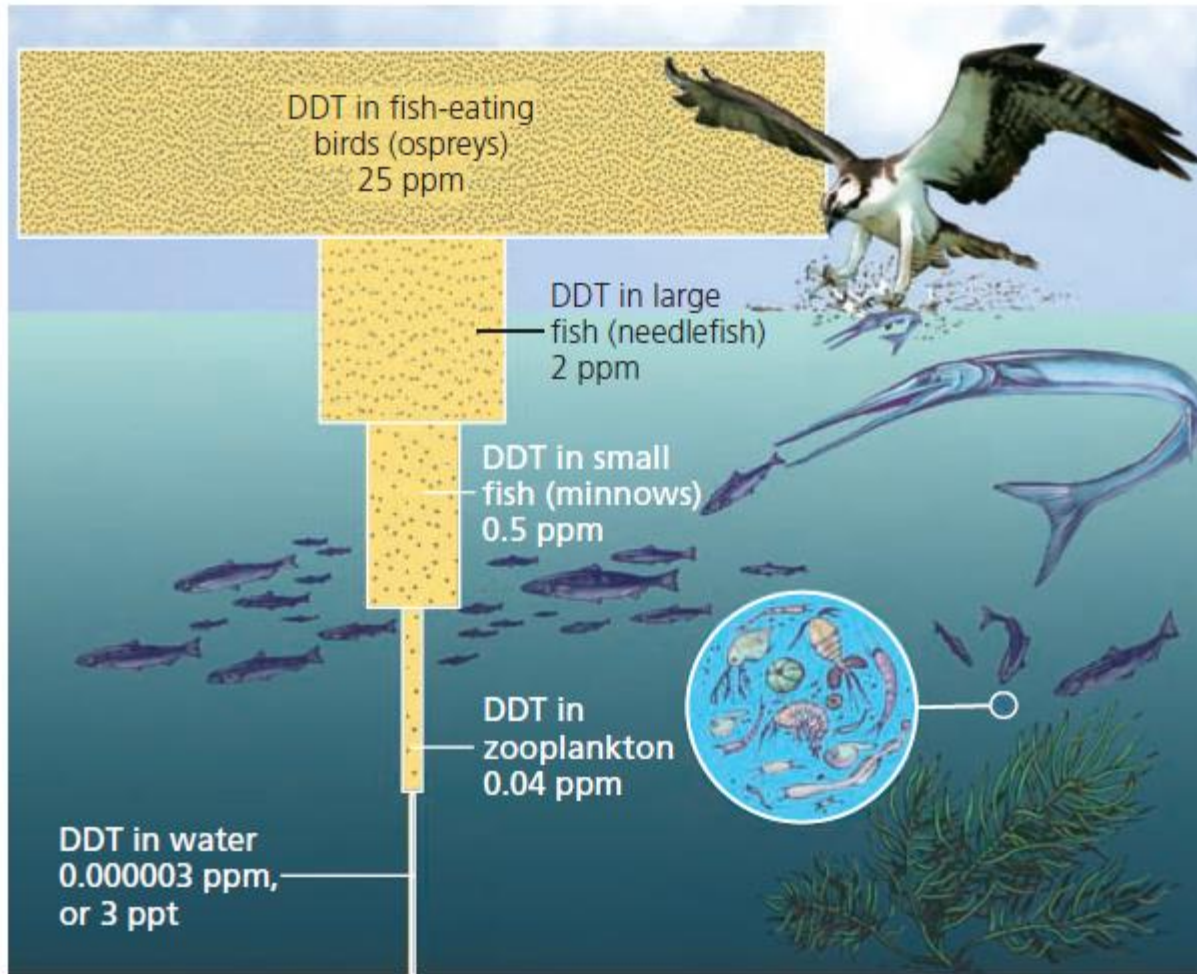
- Pollution also threatens some species with extinction as has been shown by the unintended effects of certain pesticides.

Examples:

- pesticides kill about one fifth of the honeybee colonies that pollinate almost a third of U.S. food crops
- They also kill more than 67 million birds and 6-14 million fish each year, and they threaten about one-fifth of the country's endangered and threatened species.
- During the 1950s and 1960s, populations of fish eating birds such as ospreys, brown pelicans, and bald eagles plummeted. A chemical derived from the pesticide DDT, entered surface waters

Threats to biodiversity

- Pollution



Threats to biodiversity

- Climate change

projected *climate change* could help to drive a quarter to half of all land animals and plants to extinction by the end of this century.

Examples:

- Scientific studies indicate that **polar bears** and 10 of the world's 17 **penguin** species are already threatened because of higher temperatures and melting sea ice in their polar habitats.

Threats to biodiversity

- Climate change

projected *climate change* could help to drive a quarter to half of all land animals and plants to extinction by the end of this century.



Threats to biodiversity

- Illegally Killing, Capturing and Selling of Wild Species

Some protected species are illegally killed (poached) for their valuable parts or are sold live to collectors.

- the Indian or Bengal tiger is at risk because a coat made from its fur can sell for as much as \$100,000 in Tokyo.



Threats to biodiversity

- Illegally Killing, Capturing and Selling of Wild Species

Some exotic plants, especially orchids and cacti are endangered because they are gathered (often illegally) and sold to collectors to decorate houses, offices, and landscapes.



Threats to biodiversity

- Illegally Killing, Capturing and Selling of Wild Species

Overexploitation of birds and other species is a major threat to bird populations

- Fifty-two of the world's 388 parrot species are threatened partly because so many parrots are captured (often illegally) for the pet trade.



Threats to biodiversity

- **Overfishing**

- Modern industrial fishing has caused as much as 80% depletion of some wild fish species in only 10-15 years.
- Overfishing usually results in only a temporary depletion of fish stocks, as long as depleted areas and fisheries are allowed to recover.



Conservation of Biodiversity

Conservation of Biodiversity

Some of the remedial measures mentioned below will be useful for conservation and restoration of the biotic resources

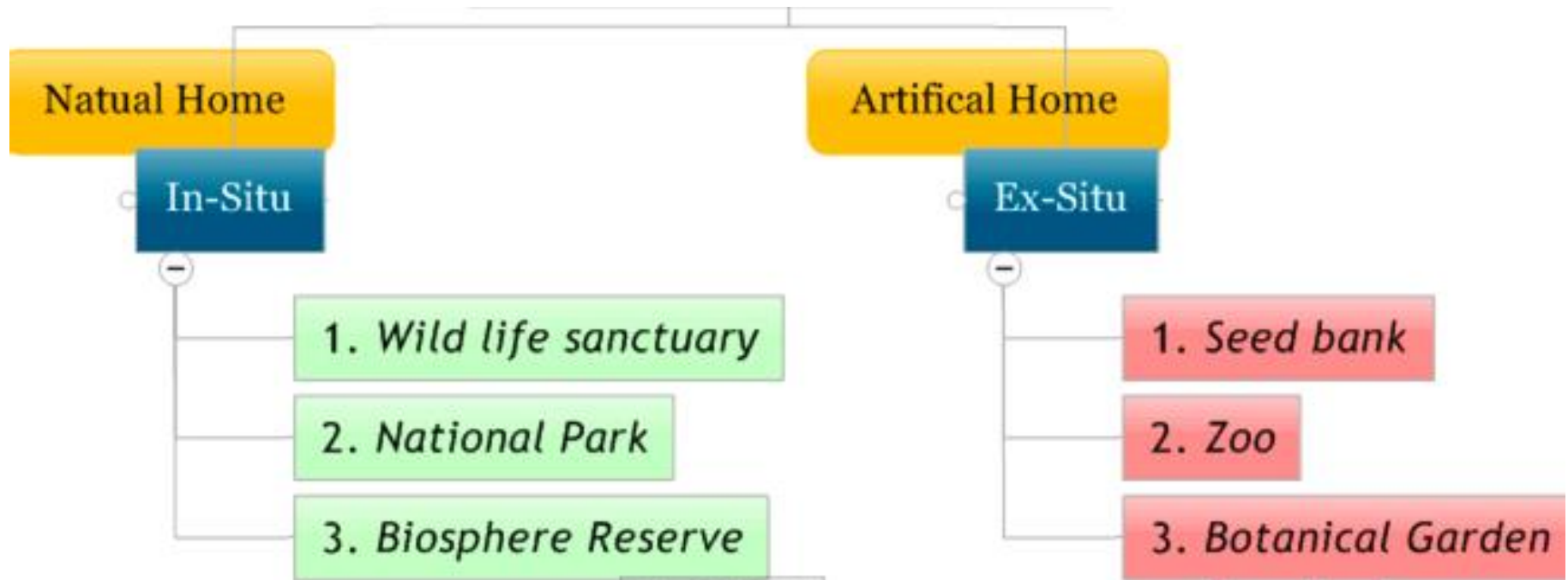
- **Assessment** of the biodiversity inventory
- Wildlife conservation measures focussing on **protecting** animal and plant life in zoos, sanctuaries, national parks and gardens.
- Setting up of **seed bank and gene bank**
- Modification of agricultural practices encouraging mixed cropping and poly culture

- ***Restoration*** of habitats and eco-systems which are important for biological communities (Reforestation, prevention of soil erosion, fencing, fertilisation).
- Imparting environmental ***education and awareness*** and motivate communities to conserve resources
- Introduction and implementation of stringent legislations
- ***Population control*** and check on indiscriminate urbanization.

Conservation Approach

Two basic approaches of conservation

- **In situ conservation**: Conserving animals and plants in their natural habitats. Large areas are kept reserve for wild life
- **Ex situ conservation**: Conservation is done under human care. Endangered plants and animals are collected and are cared for and reproduced in controlled conditions.



In Situ Conservation

Wildlife Sanctuary	Biosphere Reserves	National Parks
It is a natural habitat, owned by the government or private agency, that safeguards a particular species of birds and animals	A reserved area of land established by the government to protect the environment as a whole	A national park is a reserved area of land owned by the government which is protected from industrialization, human exploitation, and pollution. National Parks may cover Sanctuaries, and reserves as well.
Animals, birds, insects, reptiles, etc	Meant to preserve the biodiversity of a specified area	Flora, fauna, landscape, historical objects, etc

In Situ Conservation in India

- There are **18** biosphere reserves, **103** national parks and **543** wildlife sanctuaries in India.
- **Biosphere Reserves**: Protects larger areas of natural habitat, than a national park or animal sanctuary and often include one or more national parks
- **National Park** is an area dedicated for the conservation of wildlife along with its environment. Each National Park usually aims at conservation specifically of some particular species of wildlife along with others.
- **Wildlife Sanctuaries**: also protects the wildlife from killing, hunting, shooting etc., by humans.

Ex Situ Conservation

- This type is mainly for conservation of crop varieties and all the local varieties with main objective of conserving the total genetic variability of the crop species for ***future crop improvement***.

Important seed/gene bank in india are

- National Bureau of Plant Genetic Resources (**NBPGR**) in New Delhi
- National Bureau of Animal Genetic Resources (**NBAGR**) in Karnal, Haryana
- National Facility for Plant Tissue Culture Repository (**NFPTCR**) created within NBPGR.
- There are about 40 Botanical gardens available in India

Ex Situ Conservation

Zoological parks	Botanical gardens
It is a place where wild animals are kept under human observation in the natural environment.	A botanical garden is defined as that place, where different varieties of plants are cultivated.
The main purpose of setting zoological parks are to maintain ecological balance in the environment and conserve wildlife.	Botanical gardens are established for conservation of flora and scientific study.

How can we protect wild species from extinction?

What Can You Do?

Protecting Species

- Do not buy furs, ivory products, or other items made from endangered or threatened animal species.
- Do not buy wood or paper products produced by cutting old-growth forests in the tropics.
- Do not buy birds, snakes, turtles, tropical fish, and other animals that are taken from the wild.
- Do not buy orchids, cacti, or other plants that are taken from the wild.
- Spread the word. Talk to your friends and relatives about this problem and what they can do about it.